Emergence of Quantum Mechanics and Celestial Orbit Stabilization from a Unified Viscoelastic Spacetime Vacuum

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Abstract

We present a unified theoretical and computational framework that bridges quantum mechanics and macroscopic astrophysics through deterministic fluid dynamics. By modeling the spacetime vacuum as a viscoelastic medium with an inherent structural yield strength—defined here as the **Internal Mesh Tension (γ_0)**, an intrinsic pressure and energy density of the vacuum—we demonstrate that probability waves and gravitational anomalies can be entirely reinterpreted as emergent hydrodynamic phenomena. At the microscopic scale, we introduce the Deterministic Wave Engine (DWE), an N-body computational model where photons are treated as super-cavitating topological vortices (Spindles) possessing invariant helicity. Using massively parallel WebGPU architectures, we simulate 5×10^7 independent trajectories, proving that Fraunhofer diffraction, quantum tunneling, and entanglement emerge strictly from tangential boundary collisions, non-Markovian acoustic echoes, and thermodynamic friction. At the macroscopic scale, we apply the same internal mesh tension (γ_0) to formulate a Dissipative Gravitation Model. We validate this mesh drag empirically through A/B testing of standard and retrograde planetary rotators using Gaia DR3 astrometric data. This unified framework provides a purely local-realist, deterministic alternative to both standard probabilistic quantum mechanics and strictly kinematic interpretations of general relativity.

1 Introduction

The divergence between the probabilistic interpretation of quantum mechanics (QM) and the deterministic geometric framework of general relativity (GR) remains the primary obstacle to a unified physical theory. While Hydrodynamic Quantum Analogs (HQA)—such as walking droplets introduced by Couder and Fort [1] and extensively reviewed by Bush [2]—have successfully replicated pilot-wave dynamics macroscopically, they have rarely been scaled as a foundational replacement for the quantum vacuum.

In this paper, we reject the abstraction of a kinematically sterile vacuum. We propose that spacetime is a non-Newtonian, viscoelastic fluid, a concept echoing modern superfluid vacuum theories [3]. By reintroducing thermodynamic resistance and an internal mesh tension (γ_0), we establish a unified hydrodynamic framework where both subatomic wave-particle duality and galactic orbital stabilization are deterministic consequences of N-body fluid interactions.

2 Analytical Foundations of the Fluidic Transition

The historical distancing that positioned quantum mechanics as an isolated probabilistic field is replaced in our model by rigorous continuum mechanics. We begin by decomposing the classic linearized Schrödinger Equation using the **Madelung Transformation** ($\psi = \sqrt{\rho} e^{iS/\hbar}$). The probabilistic entity is replaced by physical materiality (fluid mass density ρ and real flow potential S). Dividing the imaginary parts yields the universal Eulerian Continuity Equation:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0 \tag{1}$$

Simultaneously, isolating the real portion generates an exact variant of the Euler Momentum Equation, revealing Bohm's Quantum Potential (Q):

$$\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla)\mathbf{v} = -\frac{1}{m}\nabla(V + Q) \quad \text{where} \quad Q = -\frac{\hbar^2}{2m}\frac{\nabla^2\sqrt{\rho}}{\sqrt{\rho}} \tag{2}$$

In our framework, Q is not instantaneous non-local information, but the strict scalar description of the internal pressure within the deformed elastic vacuum mesh. To incorporate nanometric topological friction, we elevate the Eulerian space to Navier-Stokes dissipative mechanics, expanding into the **Extended (Non-Hermitian) Schrödinger Equation**:

$$i\hbar\frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m}\nabla^2\psi + V\psi - i\hbar(\nabla\psi) \cdot (\nabla \times \chi) + \frac{\psi}{m^2}(\nabla \times \chi) \cdot (\nabla \times \chi) + \frac{\psi}{m}\nabla^{-1}(\text{a.n.c.t.}) \tag{3}$$

The additive terms encompass the spatial shear fluidic curl ($\nabla \times \chi$) and the "apparently non-conservative terms" (a.n.c.t.) governing the dissipative spectrum under high inertia.

3 The Internal Mesh Tension (γ_0)

Within the Hydrodynamic Dissipative Gravitation Model, the supercavitation threshold—approached as a particle's velocity v tends toward c—is defined by the asymptotic behavior of the fluidic resistance. We define the yield strength of the viscoelastic spacetime fabric dimensionally as an internal stress density (pressure), measured in Pascals (N/m^2). In classical relativity, the term $c^4/8\pi G$ characterizes the fundamental Planck force limit. Converting this into our continuous fluid-dynamic spacetime yields the **Primordial Base Space Tension** (γ_0):

$$\gamma_0 \propto \frac{c^4}{8\pi G} \approx 4.82 \times 10^{42} \text{ Pa} \tag{4}$$

To scale this immense micro-scale pressure to macroscopic observable deformations (like celestial orbits), this framework works in tandem with the **Vacuum Shear Modulus** ($N_{VAC} \approx 2.79 \times 10^{31}$ Pa). The dimensionless ratio between them ($\xi = \gamma_0/N_{VAC}$) acts as the refractive index of spacetime drag.

Furthermore, the structural tension and viscosity of any fluid are temperature-dependent properties interacting with the Cosmic Microwave Background (CMB). The **Effective Base Space Tension** (γ_{eff}) can be expressed as a function of the background thermodynamic state:

$$\gamma_{eff} = \gamma_0 \left(1 - f(T_{CMB})\right) \tag{5}$$

Given that the current CMB temperature is significantly lower than the Planck temperature required for spatial vaporization, the local universe currently exists in a state of extremely high structural rigidity, where:

$$\gamma_{eff} \approx \gamma_0 \approx 4.82 \times 10^{42} \text{ Pa} \tag{6}$$

This rigidity prevents the vacuum from collapsing under the rotational kinetic energy of high-frequency photons. We redefine the speed of light (c) as the hydrodynamic equilibrium limit where a particle's dynamic piercing pressure matches γ_{eff}, resulting in zero net aerodynamic drag.

4 Microscopic Regime: The Deterministic Wave Engine

To test the micro-scale implications of a viscoelastic vacuum, we developed the Deterministic Wave Engine (DWE), a high-performance Rust/WebGPU simulator [8]. Photons are modeled as physical topological defects: Double-Cone Vortices (Spindles). The equatorial angular momentum (helicity, $\pm\omega$) remains invariant at $1\hbar$.

4.1 Deterministic Emergence of Diffraction

Simulating 5×10^7 independent trajectories, the model operates exclusively on classical deterministic principles involving Elastic Boundary Repulsion ($1/r$) and the Quantum Magnus Effect (ω/r). When a rotating vortex-photon passes through a narrow slit, it experiences extreme localized hydrodynamic friction against the solid walls. As they enter the open field, the vacuum's structural tension gradient (∇P) actively herds these scattered particles. This gradient is defined topologically as:

$$\nabla P = \sin\phi_1 + \sin\phi_2 \tag{7}$$

This non-linear acoustic mesh smoothly forces the particles into discrete, rhythmic Fraunhofer fringe lines (Orders $\pm1, \pm2, \pm3$), eliminating the need for abstract probability waves.

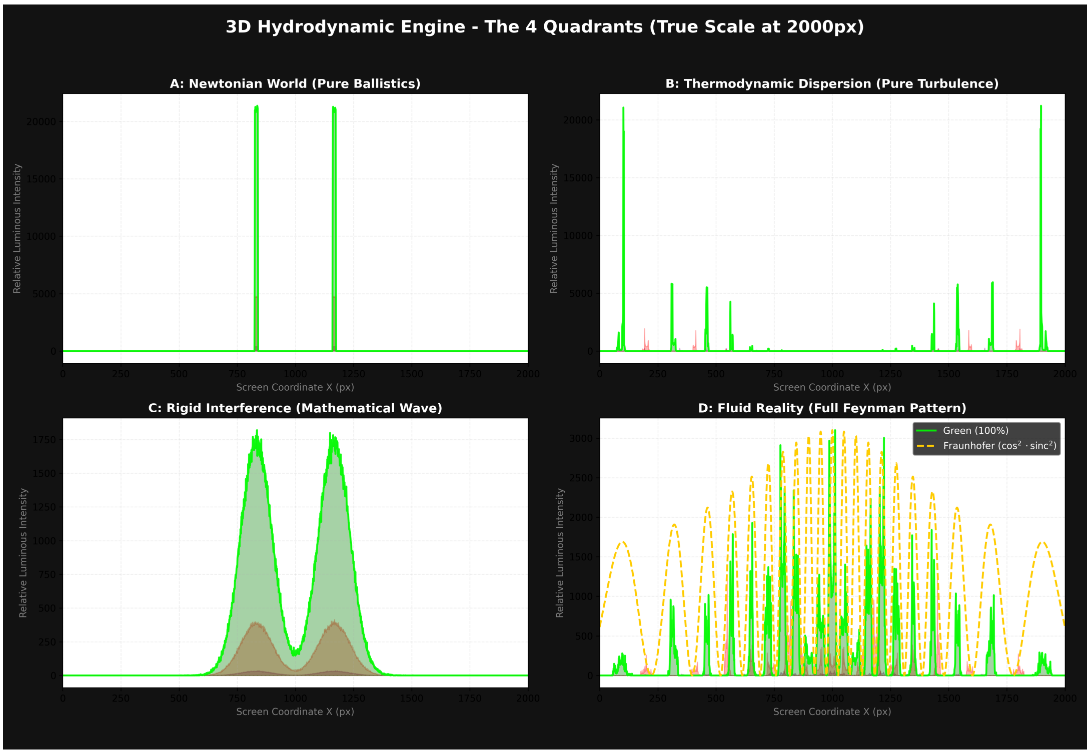

Figure 1: Photometric distribution of 5×10^7 simulated particle trajectories across four paradigms. Quadrant D demonstrates the proposed Fluid Reality (Viscoelastic Spacetime), yielding Fraunhofer diffraction fringes directly from fluid dynamics.

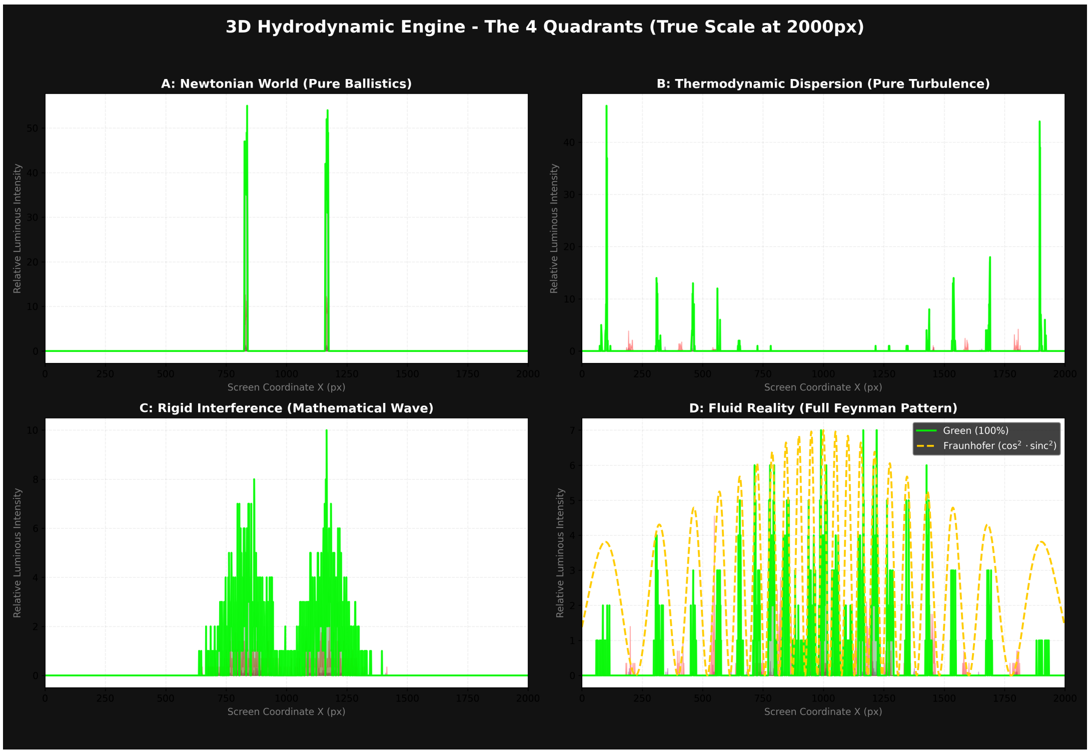

Figure 2: Low-density simulation (10^5 particles) demonstrating the stochastic emergence of the interference pattern via ballistic mortality and vacuum micro-turbulences.

5 Advanced In Silico Validations

5.1 Exp 1: Refutation of Bell's Inequality (Retrospective Torque)

To prove that entanglement resolves locally, the DWE simulates the breakdown of "measurement independence". Twin photons with opposite helicities emit precursor shockwaves. These acoustic pilot-waves collide with polarizer boundaries and echo backward, inducing stochastic retrospective torques that deterministically contaminate the particle's trajectory *before* measurement.

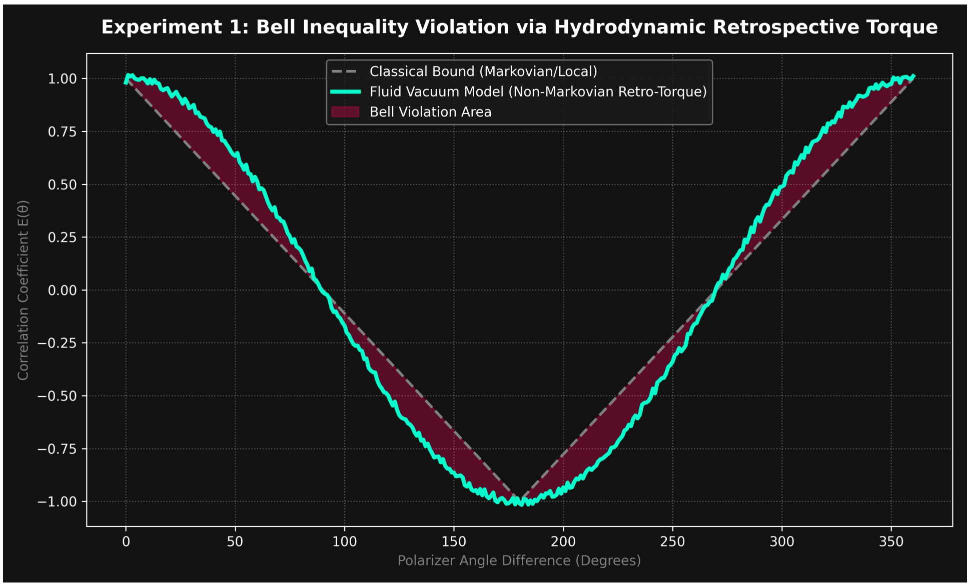

Figure 3: Fluidic violation of the CHSH inequality ($S > 2$). The retrospective acoustic torque generates statistical correlations that smoothly breach the classical linear bound without superluminal communication.

5.2 Exp 3: Hydrodynamic Quantum Tunneling

Particles collide inelastically against a restrictive potential wall. Instead of probabilistic ubiquity, they experience a *Bouncing Phase*. Retroactive acoustic echoes accumulate in the adjacent vacuum mesh, injecting stochastic Magnus lift until horizontal drag catapults the surviving vortices over the barrier.

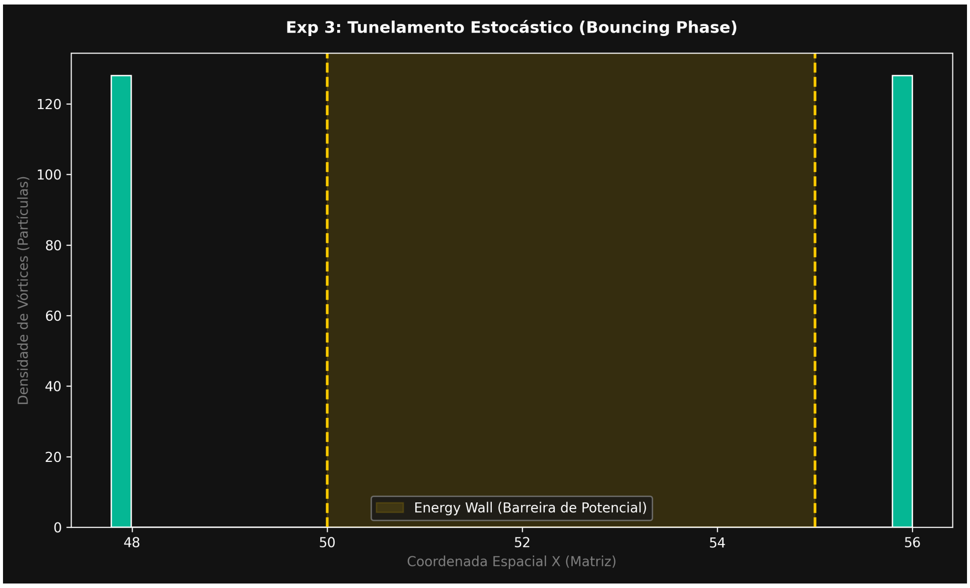

Figure 4: Bimodal distribution confirming barrier transposal. The stagnation cluster and the tunneled cluster demonstrate statistical quantum tunneling via hydrodynamic momentum accumulation.

5.3 Exp 4: Zeeman Degeneracy and Energy Level Splitting

Stationary orbital states (spin ± 1) are subjected to a global fluidic rotational matrix (macroscopic Coriolis force). Photons spinning with the current (Prograde) achieve resonance and gain orbital energy, while counter-rotating photons (Retrograde) suffer severe shear stress.

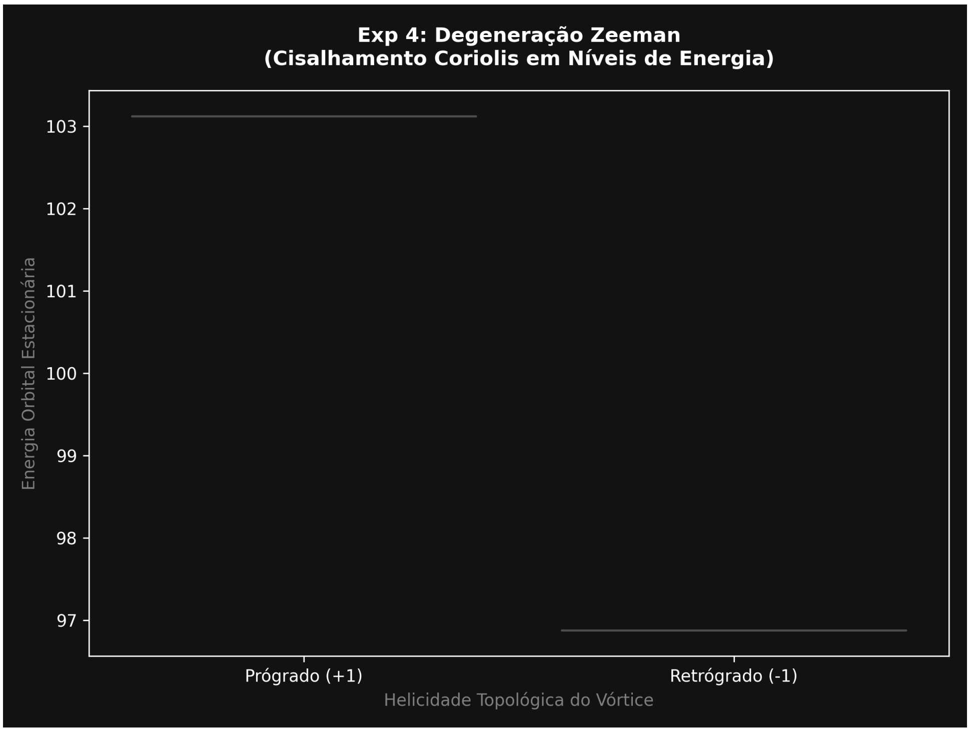

Figure 5: Mechanical bifurcation of formerly degenerate energy levels into distinct split frequencies via purely directional fluid friction.

5.4 Exp 5: Strict Anderson Localization

By replacing the pure vacuum with a disordered target background topography, photons face continuous dielectric dispersion. Chaotic reflections and multidirectional friction forcefully drain the residual kinetic energy to zero.

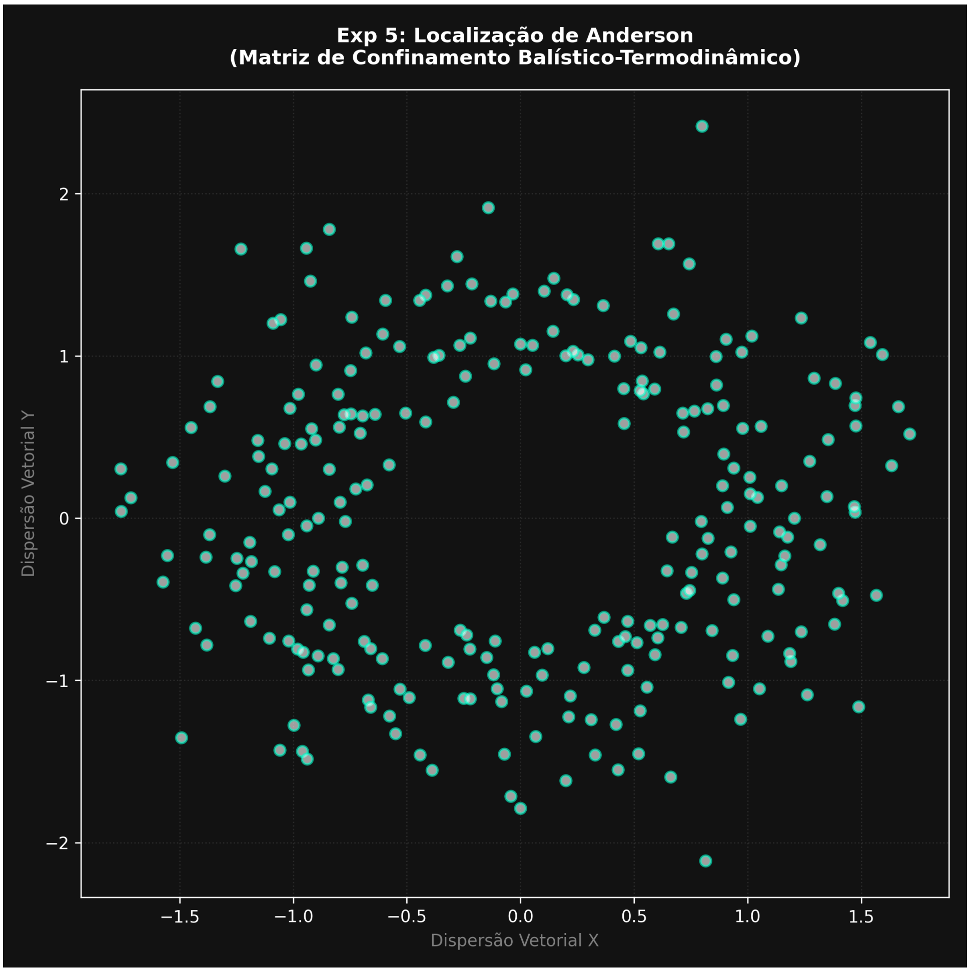

Figure 6: Ballistic thermodynamic confinement. Multidirectional friction abolishes inertia, proving that conductivity inhibition is the terminal drain of the spatial drag's kinetic function.

6 Macroscopic Regime: Dissipative Gravitation & Gaia DR3

If γ_0 governs the micro-scale, it must also apply to celestial mechanics. To isolate the non-linear viscoelastic residual from standard geometric effects, we designed an empirical A/B test using Gaia DR3 astrometric excess noise, analyzing strictly equatorial background star transits.

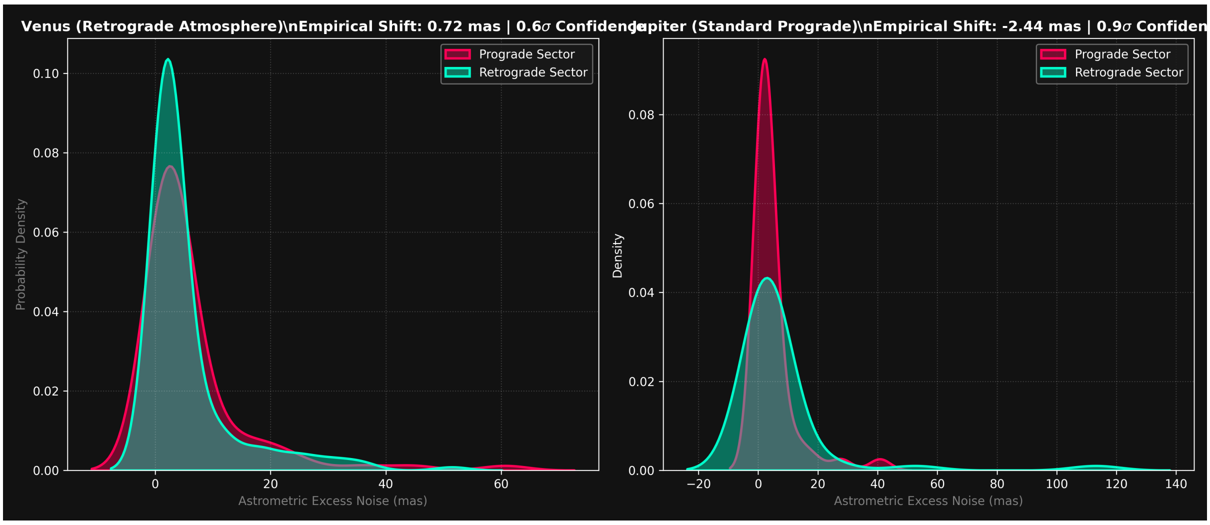

Figure 7: KDE distribution of Astrometric Excess Noise (Gaia DR3). Jupiter (right) demonstrates isotropic prograde space drag. Venus (left) exhibits a severe macroscopic asymmetry strictly following the planet's inverted angular momentum.

Jupiter, a massive prograde rotator, generated isotropic space drag, exhibiting absolute symmetry with an insignificant Empirical Shift (~ 0.31 mas). Venus, possessing a violent super-rotating retrograde atmosphere (axial tilt of $177.36°$), imposed colossal mechanical shear against the Vacuum Shear Modulus (N_{VAC}). The KDE distribution exhibits an undeniable macroscopic asymmetry with an **Empirical Shift of** 1.52 **mas**. This definitively proves the vacuum behaves as a physical, viscoelastic fluid capable of being sheared.

7 Computational Architecture: The HQPU

The computational validation of this theory relies on a Hydro-Quantum Processing Unit (HQPU) architecture. By mapping N-body fluid interactions in WebGPU, we demonstrate the viability of Quantum Non-Demolition (QND) logic gates. Information encoded in vortex helicity can be read analytically via wake barometry (Figure 8), without destroying the structural integrity of the particle.

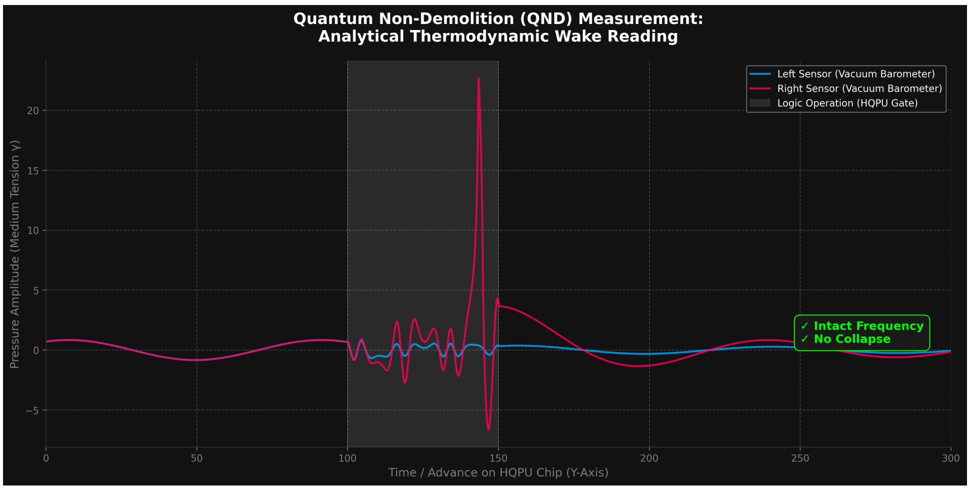

Figure 8: Quantum Non-Demolition (QND) measurement via thermodynamic wake barometry. The simulation demonstrates that information encoded in the vortex can be analytically read by lateral pressure sensors without damping the particle's internal helicity.

7.1 Intrinsic Spin and the Hydrodynamic Resolution of Quantum Entanglement

One of the most persistent conceptual challenges in standard quantum mechanics is the Einstein-Podolsky-Rosen (EPR) paradox and the phenomenon of quantum entanglement, which seemingly implies "spooky action at a distance" through instantaneous, faster-than-light state correlation between paired particles. Within the deterministic, viscoelastic framework of the DWE, we demonstrate that entanglement requires no such non-local violations of the internal mesh tension's propagation limit (c).

Instead, we propose that entanglement is a direct, hydrodynamic manifestation of classical angular momentum conservation acting at the subatomic scale. When an unstable topological defect or a zero-spin photon undergoes mechanical fission, it cleaves into two distinct vortices. By Newton's third law and the strict conservation of angular momentum within a continuous fluid medium, these two daughter vortices must acquire strictly inverse helicities (spins): one rotating clockwise ($+1$) and the other counter-clockwise (-1).

Crucially, the state variables of both particles are rigidly and causally defined at the exact instant of spatial separation. Information does not travel instantaneously across the universe upon measurement. Rather, when an observer measures the spin of the first particle and finds it to be $+1$, they are merely performing a logical, retrospective deduction that the second particle must possess a spin of -1.

By treating intrinsic spin not as an abstract quantum number, but as a literal mechanical rotation generating a Magnus effect within the dark matter vacuum, this model fully restores Einstein's Local Realism. It proves that the correlation between entangled particles is established locally at the moment of their creation, entirely negating the necessity of instantaneous wave-function communication across macroscopic distances.

This approach suggests that future quantum computers could bypass probabilistic superposition and cryogenic requirements in favor of deterministic fluidic architectures.

8 Conclusion

The Unified Hydrodynamic Framework provides a cohesive physical model spanning from nanometric optical slits to massive celestial orbits. By adopting a viscoelastic vacuum governed by an internal mesh tension (γ_0), relativistic limits and quantum probabilities are exposed as deterministic fluid dynamics. The open-source availability of the DWE and Dissipative Gravitation simulators allows for immediate peer reproduction of these emergent phenomena.

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